

WORKING CAPITAL GUARDIAN

KARBHARI

कारभारी · the one who runs the enterprise

Find what your numbers are hiding.

An agentic investigator for Indian MSMEs. It reads a business's bank-facility evidence, reconstructs its working-capital position from first principles, and shows, with citations, where Drawing Power is being lost and where it isn't.

₹12,10,000

DRAWING-POWER GAP SURFACED IN A VERIFIED LIVE RUN

BUILT FOR INDIAN MSMEs
READ · CROSS-CHECK · PROVE

AT A GLANCE

An investigator, not a chatbot.

Most Indian MSMEs run on a bank cash-credit facility whose usable limit, **Drawing Power**, is recalculated from stock, debtors and creditors every month. When those numbers are stale, mis-aged, duplicated or simply mis-read, the business quietly loses access to capital it is entitled to under its own sanction terms. Nobody inside a small enterprise has the time or specialist skill to check. KARBHARI does.

₹12.10L

capacity gap found and matched to hand-computed ground truth, to the rupee

11

tool calls the agent chose, across a 7-document evidence bundle

0

headline figures computed by the language model; Python owns the arithmetic

52

automated tests, including a live end-to-end run against the real model

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In one minute

- **What it does.** Takes a case (sanction letter, stock statements, debtor invoices and receipts, creditor ledger, bank statement) and determines whether the bank is recognising the Drawing Power those documents actually support.
- **How it does it.** A bounded agent loop decides what to read and which checks to run. Every number comes from deterministic, tested Python: debtor reconciliation, consistency checks and the Drawing Power formula.
- **What you get.** Graded findings (**SUPPORTED** **UNRESOLVED** **CONTRADICTED**), each with evidence quotes, plus the complete trace of how the agent got there.
- **Why it is different.** It is built to be right, not agreeable. It reports what undermines the owner's case as readily as what supports it.

01 • WHY KARBHARI EXISTS

Large companies have departments. MSMEs have the owner.

Big enterprises do not win only because of software. They win because they employ specialists in every domain (treasury, credit, collections, compliance, analytics) who understand problems that cut across all of them. The owner of a small Indian enterprise has to be all of those people at once.

The specialist-capability gap

An MSME may have accounting software, a bank portal and a part-time accountant. What it lacks is someone who can look across all of them, understand what a problem actually involves, and act on it. The real challenge is not "*MSMEs need better software*". It is that small businesses lack the specialist capacity to understand and coordinate complex, cross-functional problems.

KARBHARI's long-term vision is to be that missing operator: a **problem-first** layer where the owner brings a real situation in their own words, and does not need to know which department, professional or software category it belongs to.

Why working capital first

Working capital is where this gap hurts most and is easiest to prove. A cash-credit facility's usable limit moves every month with the stock statement and the debtor book. If the documents behind it are inconsistent, the owner sees only a smaller number at the bank, with no way of knowing whether it is right.

WHERE THE NUMBER GOES WRONG

Five ways Drawing Power is mis-stated

1. **Stale or swinging stock statements.** A large month-on-month swing, or a statement older than the facility allows, makes the stock figure unreliable.
2. **Debtors aged incorrectly.** Invoices past the sanctioned cut-off (e.g. 90 days) must be excluded; valid ones must not be.
3. **Duplicate ledger entries.** The same invoice entered twice inflates the debtor book.
4. **Receipts that don't match invoices.** Untagged payments, partial payments and over-payments distort what is truly outstanding.
5. **Assumed rather than sanctioned terms.** Margins and creditor deductions that differ from the sanction letter change the answer.

Errors run in **both directions**. KARBHARI is not built to argue for a bigger number; it is built to find the right one, and to say clearly when the evidence does not support a claim.

Under RBI's IRACP norms, Drawing Power based on a stock statement more than three months old is irregular; if the irregularity continues for 90 days the account must be classified NPA. KARBHARI states this rule whenever staleness is relevant.

02 • WHAT KARBHARI IS

A business operator that investigates before it answers.

KARBHARI is a problem-first agentic product for Indian MSMEs. Its first fully built capability, the **Working Capital Guardian**, works out whether a business's Drawing Power is being correctly recognised, and proves it.

UNDERSTAND

Reads the whole case

Sanction terms, stock statements, ledgers, invoices, receipts and bank statements, in PDF, Excel, CSV or text.

REASON AND VERIFY

Rebuilds the numbers

Reconstructs eligible debtors invoice by invoice, checks periods against each other, and computes Drawing Power under the sanction's own terms.

DELIVER

Shows its working

Graded findings with quoted evidence, the money at stake, and the full step-by-step trace of the investigation.

Not a chatbot, by design

QUESTION	GENERIC AI ASSISTANT	KARBHARI
Starting point	A question about business in general	A specific case with its own documents
Who does the maths	The language model, in prose	Deterministic, tested Python; the model never states the headline figure
Debtor figure	Copied from a ledger's subtotal line	Independently rebuilt from invoices and receipts
Process	One-shot answer	Bounded investigation: the agent chooses which documents and tools, up to 20 steps
Output	Advice ("improve your credit terms")	Graded findings, amounts and evidence quotes
Unfavourable facts	Often smoothed over	Actively sought and reported as unresolved or contradicted
Auditability	None	Every thought, tool input and tool output logged and shown in the UI

WHO IT IS FOR

- Small manufacturers, traders and distributors with a stock-and-debtor cash-credit facility
- Family-run enterprises without a treasury or credit team
- Accountants checking a client's Drawing Power

WHAT IT WILL NEVER CLAIM

- What a bank *will* do, or that money is legally owed
- Any number that a tool did not produce
- More credit than sanctioned: the limit is always the hard ceiling

03 • DRAWING POWER, EXPLAINED

The number your bank uses, and how KARBHARI rebuilds it.

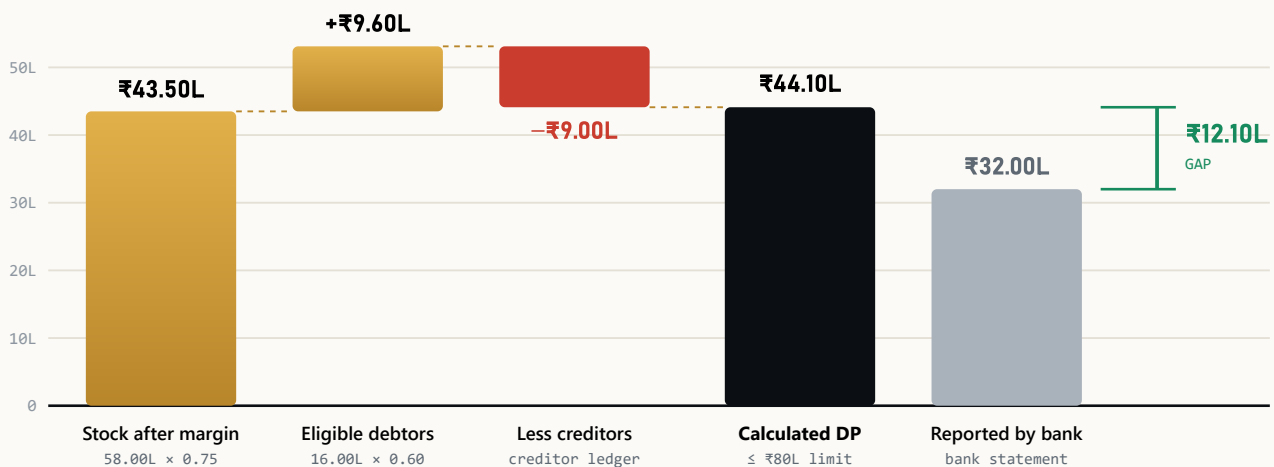
A cash-credit facility is sanctioned up to a limit, but the business can only draw what its current collateral supports. That figure is Drawing Power. KARBHARI computes it in exactly one place in the codebase, `calculations/drawing_power.py`.

STOCK AND BOOK-DEBT HYPOTHECATION • STANDARD FORMULA

$$DP = \text{Stock} \times (1 - \text{stock margin}) + \text{Eligible debtors} \times (1 - \text{debtor margin}) - \text{Creditors}$$

...then capped between ₹0 and the sanctioned limit. Gaps under ₹1,000 are treated as rounding noise, not findings.

Worked example: the verified Suryoday Textiles case



Every gap in the inputs is said out loud

If an input is missing, the engine does not quietly guess. Each fallback is written into `assumptions_used` and carried into the finding:

- No sanctioned margins found → typical 25% / 40% used, and flagged as a placeholder to confirm.
- No ageing breakdown → total debtors used, and flagged as an *upper bound*.
- No creditor figure → treated as zero, and flagged as *optimistic*.
- Collateral above the limit → capped, and the cap is recorded.

Any assumption downgrades the verdict

A positive gap is graded **SUPPORTED** only when no assumptions were needed. Otherwise it is **UNRESOLVED**.

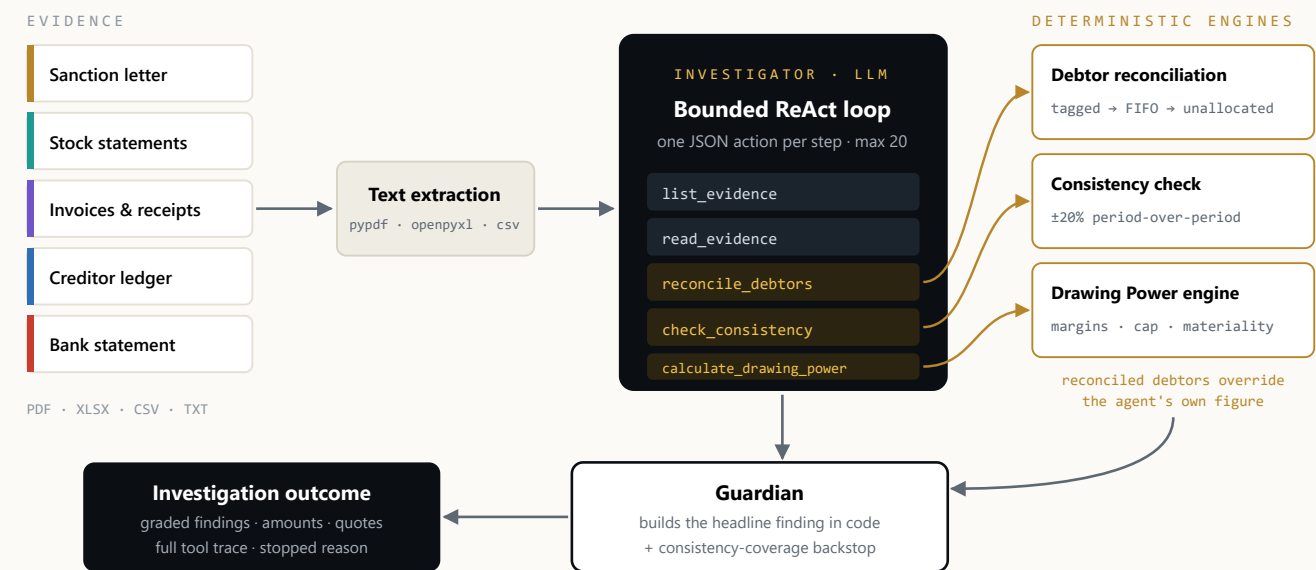
If the calculated figure is at or below what the bank recognises, the finding is **CONTRADICTED**: the evidence does not support a claim of lost capacity.

The gap is framed as headroom **within the existing sanctioned limit**: never a request for more credit, and never a guarantee the bank will release it.

04 • HOW IT WORKS

The model reasons. Python calculates.

KARBHARI separates judgement from arithmetic. A language model does what it is good at: reading messy documents, deciding what to look at next, and structuring line items. Every number that matters is produced by deterministic code that cannot hallucinate.



ENFORCED IN CODE • 1

Headline built by the system

The Drawing Power gap finding is constructed by `guardian.py` straight from the engine's return value. The agent is told not to restate it and has no way to alter it.

ENFORCED IN CODE • 2

Reconciled debtors win

If debtors were reconciled, `calculate_drawing_power` overrides whatever eligible-debtor figure the agent passes in, and records the override where it can be seen.

ENFORCED IN CODE • 3

Per-debtor isolation

Reconciliation groups records by debtor before matching, so one customer's payment can never settle another customer's invoice, whatever the agent sends.

05 • THE INVESTIGATION LOOP

A real investigation, not a fixed pipeline.

Each turn, the model sees the running transcript and returns exactly one action: call a tool, or finish. It decides which documents to read, in what order, and when it has enough. A required checklist keeps it thorough; the step limit keeps it bounded.

1

list_evidence**Inventory the case**

See every document attached: ID, filename and category.

2

read_evidence × every item**Read everything, no exceptions**

The agent must read every item, even ones that look redundant. A document it did not read cannot inform its conclusion. Figures it uses must be traceable to a quote it actually saw.

3

check_consistency**Compare periods**

If two documents or periods report the same field (two stock statements a month apart is the classic case), flag any swing of 20% or more for a human to explain.

4

reconcile_debtors**Rebuild the debtor book**

When invoice- or receipt-level data exists, never trust a ledger's subtotal. Reconstruct eligible debtors from line items, as of the evidence's own date.

5

calculate_drawing_power**Compute, or explain why not**

Run the engine with the facility terms and figures found. If inputs are missing, it says exactly which ones.

6

action: finish**Report findings**

Everything except the headline gap, which the system adds itself: data-quality issues, ambiguities, and anything that cuts against a favourable conclusion.

ONE STEP, AS THE MODEL RETURNS IT

```
{
  "thought": "Two stock statements cover
  Aug and Sep. Compare them
  before trusting Sep alone.",
  "action": "call_tool",
  "tool": "check_consistency",
  "tool_input": {
    "field_history": {
      "stock_value": [
        {"period_label": "Aug-26",
        "value": 4200000},
        {"period_label": "Sep-26",
        "value": 5800000}
      ]
    }
  },
  "findings": null
}
```

→ 1 variance flag: stock value +38.1%

LOOP MECHANICS

- **Bounded:** 20 tool calls maximum, so cost and runtime stay predictable.
- **Resilient:** one shared retry budget absorbs API errors and malformed or corrupted JSON, both observed in real runs.
- **Forgiving where safe:** if the model mistakenly "calls" `finish_investigation` as a tool, it is treated as a finish, not a wasted step.
- **Recorded:** every step's thought, tool, input and output is saved as the trace.
- **Honest stop reasons:** complete, incomplete (step limit or invalid action), or error.

06 • FEATURES

What it does, and why.

Case and evidence workspace 01

Create a case per business, drop in evidence and tag each file with a category. A completeness checklist shows what is still missing.

- Reads PDF, XLSX, CSV and plain text
- Each document bounded to 20,000 characters for predictable prompts
- Scanned PDFs without a text layer are flagged plainly and never crash the run

WHY Real MSME records arrive in every format. A bad file becomes a visible gap, not a broken investigation.

Bounded agentic investigator 02

A ReAct-style loop in which the model chooses which documents to read, which checks to run, and when to stop, up to 20 steps.

- Required checklist: read everything, compare periods, rebuild debtors, compute
- Single JSON action protocol with retries
- Provider-agnostic LLM client (Gemini today)

WHY Every case is shaped differently. A fixed pipeline breaks on the second case; an unbounded agent is expensive and unpredictable.

Deterministic debtor reconciliation 03

Rebuilds outstanding and *eligible* debtors from invoices and receipts, per debtor, as of the evidence's own date.

- Three-tier matching: invoice-tagged → FIFO against the same debtor's oldest open invoices → explicitly unallocated
- Duplicate invoices detected and excluded, with the amount shown
- Ageing checked against the sanctioned cut-off (e.g. 90 days)

WHY A ledger subtotal is someone else's arithmetic. Tally-style ledgers rarely tag receipts, so FIFO by debtor is how a credit analyst would do it.

Period-over-period consistency 04

Flags any field (stock, debtors, creditors) that moves 20% or more between consecutive periods, and any figure that appears from zero.

- Threshold configurable per call
- Flags, never explains: the agent and the human judge the cause

WHY A big swing may be growth, seasonality or a data error. It should never pass silently into a Drawing Power figure.

Drawing Power engine 05

The single place the number is computed: sanctioned margins, eligible debtors, creditor deduction, capped at the limit, compared with the bank's figure.

- Every fallback recorded in `assumptions_used`
- Missing inputs listed by name and source document
- ₹1,000 materiality threshold

WHY Money arithmetic must be exact, repeatable and testable. Language models are none of those things.

Graded findings with citations 06

Each finding carries a status, a plain-language explanation, the rupee impact where known, evidence IDs and exact quotes.

SUPPORTED

UNRESOLVED

INELIGIBLE / CONTRADICTED

WHY An owner taking a finding to the bank needs to know how sure it is, and exactly which line of which document backs it.

"How it investigated": the full trace

07

Every step the agent took is shown as a timeline: its reasoning, the tool it chose, the exact input, the raw output and a one-line summary.

- Generated live from the actual run, not reconstructed afterwards
- Stored with the investigation for later review

WHY "Trust me" is not good enough for money. An owner, accountant or banker should be able to follow the reasoning step by step.

Honesty guardrails

08

The investigator is told to look actively for what undermines a favourable result (stale statements, duplicates, unallocated receipts, unexplained swings) and to report it.

- No numeric claim a tool didn't produce
- No claims about what a bank will do
- "Appears supportable under the supplied facility terms", never "the bank owes you"

WHY An advocacy engine would be useless in front of a credit officer. Credibility is the product.

Coverage backstop

09

If the case has several period-specific documents of the same category but the agent never ran a consistency check, the system adds an explicit

UNRESOLVED finding saying so.

- Never invents a variance; only reports the gap in coverage
- Based on a cheap deterministic signal, not a re-read

WHY Where the agent's discipline was observed to be unreliable, the gap is made visible in code instead of hoping a prompt fixes it.

Film-grade investigation console

10

A dark financial-investigation console with a persistent rail: *Findings · Investigation · Evidence · How it investigated · Actions*.

- Count-up headline, KPI strip, capacity bars, eligibility ring
- Full-screen view of the agent reading each document while it runs
- Subtle synthesized sound effects, with a mute toggle

WHY The answer is one number; the confidence comes from being able to drill into everything behind it, and enjoying doing so.

Resilient agent loop

11

If the model tries to finish before calculating Drawing Power, it is sent back once to do so. Gemini rate limits are waited out as requested instead of failing the run.

- Premature finishes deferred once, visible in the trace
- Rate-limit waits capped at 45 s, never an infinite loop

WHY A missing headline or a failed run in front of a banker costs more trust than a few extra seconds.

Packaged and tested

12

A single Docker image serves the API and the built console. 52 automated tests cover extraction, reconciliation, consistency, Drawing Power, tools, the agent loop, the API and frontend serving, plus a live model run.

- The full 7-document case verified inside the container
- Real bugs found in verification became permanent regression tests

WHY For a financial tool, "it worked on my laptop once" is not evidence. Reproducibility is.

07 • TRUST AND GUARDRAILS

Built to be right, not agreeable.

A tool that tells an owner what they want to hear is worse than no tool. KARBHARI's guardrails live in three places: the prompt, the tools, and the code that assembles the final result.

How a finding is graded

STATUS	MEANING	TYPICAL EXAMPLE
SUPPORTED	Evidence and a tool output directly back it, with no assumptions needed.	Calculated DP exceeds reported DP using sanctioned margins and reconciled debtors.
UNRESOLVED	Evidence is incomplete or ambiguous, an assumption was needed, or more documents are required.	A ₹70,000 receipt that cannot be allocated to any open invoice.
INELIGIBLE / CONTRADICTED	The evidence contradicts a favourable claim, or a sanction-term condition is not met.	An invoice aged 105 days against a 90-day eligibility cut-off.

Rules the investigator follows

- Never state a number a tool didn't produce; never invent or round a figure it didn't see.
- Actively look for what undermines a favourable conclusion, and report it.
- Never assert what a bank will do or that a figure is legally guaranteed.
- State the RBI IRACP staleness rule when a stock statement is more than three months old, as the general rule, not as something that has already happened.
- If nothing was found beyond the headline, finish with no extra findings. No filler.

Rules enforced in code

- The headline gap finding is built from the engine output, never from the agent's words.
- Reconciled eligible debtors override the agent-supplied value, and the override is shown.
- Records are segregated per debtor before any matching.
- Duplicate invoices and unallocated receipts are always reported, never silently dropped.
- Any assumption downgrades a gap from supported to unresolved.
- A skipped consistency check becomes a visible unresolved finding.

5

real issues were found by verification and documented openly: four fixed, one mitigated. They include payments from one debtor settling another's invoice, which is now prevented in code and covered by a regression test. KARBHARI's README lists what is real, what is stubbed, and what is not yet reliable.

08 · CASE STUDY

Suryoday Textiles Pvt Ltd:

₹12,10,000, to the rupee.

A deliberately difficult 7-document bundle was built, and the true answer was calculated by hand **before** the agent ran. Then KARBHARI investigated it live, on the real model, in 11 tool calls.

The evidence, with traps built in

DOCUMENT	WHAT MADE IT HARD
Sanction letter (PDF)	₹80L limit; 25% / 40% margins; 90-day debtor eligibility; 3-month staleness rule
Stock statements × 2	Aug ₹42L → Sep ₹58L: a deliberate +38.1% swing
Invoices	9 invoices across 5 debtors, including one duplicate ledger row
Receipts	4 receipts: one tagged-partial, one untagged (FIFO), one over-payment
Creditor ledger	₹9L sundry creditors
Bank statement	Reported Drawing Power ₹32L

Ground truth vs. the live run

FIGURE	BY HAND	KARBHARI
Eligible debtors	₹16,00,000	₹16,00,000
Calculated DP	₹44,10,000	₹44,10,000
Reported DP	₹32,00,000	₹32,00,000
Gap	₹12,10,000	₹12,10,000

$$\begin{aligned} & ₹58L \times 0.75 + ₹16L \times 0.60 - ₹9L \\ &= ₹43.5L + ₹9.6L - ₹9L = ₹44.10L \\ & ₹44.10L - ₹32.00L = ₹12.10L \end{aligned}$$

What the agent caught on its own

EXCLUDED

Duplicate invoice, ₹3,50,000: the same invoice entered twice; removed from the debtor book with the excluded amount shown.

FLAGGED

Unallocated receipt, ₹70,000: an over-payment with no open invoice left to apply it to.

INELIGIBLE

105-day-old invoice: past the sanctioned 90-day cut-off, so excluded from eligible debtors.

FLAGGED

Stock +38.1% in one month: caught by `check_consistency`; worth confirming before relying on September alone.

5 findings in total, split across supported, unresolved and contradicted, not one-sided. The same figures were reproduced inside the Docker image.

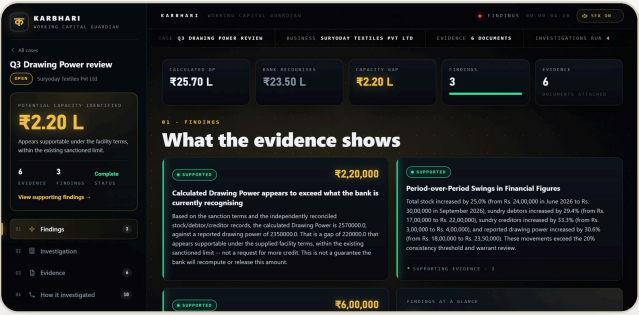
09 · THE EXPERIENCE

From a pile of documents to a defensible answer.

The owner, or their accountant, never configures a workflow. They open a case, drop in what they have, and ask KARBHARI to investigate.



INVESTIGATING · THE AGENT READS EACH REAL DOCUMENT



FINDINGS · GRADED, CITED, BACKED BY TESTED CODE

VIEW	WHAT IT SHOWS	BUILT FOR
Lobby	Open a case, the verified benchmark, a sample of the agent at work, how it works	First impression
Headline and KPI strip	The capacity gap with count-up animation, plus calculated DP, bank figure, findings and evidence	The owner's first glance
Findings	Graded findings with amounts, explanations and evidence quotes, plus a summary tile	Deciding what to raise with the bank
Investigation	Capacity bars, eligibility ring, invoice-level table, variance cards, assumptions	The accountant's check
Evidence	Drag-and-drop upload, category chips, document cards and a completeness checklist	Keeping the case complete
How it investigated	The step-by-step timeline: thought, tool, input, output	Audit and explanation
Actions	Previews of the next phase: draft note to bank, evidence pack, follow-up checklist, monthly watch	Turning findings into action

DESIGN LANGUAGE

Borrowed from KARBHARI's launch film: a near-black stage, a drifting dot grid, precise numerals and colour that always means something: gold for capacity, red for problems, green for verified. Motion and quiet sound effects make results land.

RUNS ANYWHERE

One Docker command starts the whole product: API, agent and console. Case data moves freely between Windows and the Linux container, and the six files in ASSETS FOR TESTING make a complete sample case.

10 · TECHNOLOGY AND DEPLOYMENT

Small, boring, verifiable.

A deliberately compact stack: easy to audit, quick to run, and packaged as one container.

Stack

API	Python 3.12 · FastAPI · Uvicorn
Data	SQLModel on SQLite; investigation results stored as JSON
Extraction	pypdf (PDF) · openpyxl (Excel) · csv · plain text
Model	Provider-agnostic LLMClient ; Google Gemini via REST today
Engines	Pure Python: reconciliation, consistency, Drawing Power
Console	React 19 · TypeScript · Vite · Web Audio for sound effects
Packaging	Single Docker image serving API and built console

API

GET	/api/health
POST · GET	/api/cases
GET	/api/cases/{case_id}
GET	/api/cases/{case_id}/evidence/categories
POST · GET	/api/cases/{case_id}/evidence
DELETE	/api/cases/{case_id}/evidence/{evidence_id}
POST	/api/cases/{case_id}/investigate
GET	/api/cases/{case_id}/investigations

Test coverage: 52 tests

SUITE	TESTS
Debtor reconciliation	11
Document extraction	7
Drawing Power engine	6
Consistency checks	5
Agent loop (finish guard, rate limits)	7
Agent tools (incl. the override rule)	4
Cases and evidence API	4
Frontend serving (incl. path safety)	3
Evidence storage paths (Windows / Linux)	3
Health	1
Live end-to-end run on the real model*	1
Total	52

* Skipped automatically when no API key is configured.

Run it

```
# published image
docker run -p 8000:8000 \
  --env-file backend/.env \
  yeldnever/karbhari:0.7.0

# tests
cd backend && python -m pytest -q
```

11 • LIMITATIONS AND ROADMAP

What's real today, and what comes next.

KARBHARI states its limits as plainly as its findings. These are the known gaps at v0.7.0.

Known limitations

- **The agent can still skip a check.** It may skip `check_consistency`; the code backstop makes that gap visible, but does not force the check. A premature finish before the Drawing Power calculation is sent back once.
- **Creditor-side reconciliation is simple:** one total, not invoice- and payment-level.
- **Cross-document conflicts** (two documents disagreeing on one figure) are not explicitly re-detected in the agent version.
- **One case family verified** in depth; not yet tested on 10+ documents, contradictory evidence, non-English text or illegible uploads.
- **No OCR** for scanned PDFs; **no authentication**; evidence categories are advisory.
- **Run time varies** with the model's response time, typically 30 seconds to 2 minutes.

Roadmap



Close the consistency gap

Native function-calling with enforced tool sequencing, or a stronger model tier.



Turn findings into action

The Actions tab, already previewed in the console: a draft, evidence-backed note to the bank requesting a Drawing Power recomputation.



Harder cases

Contradictory documents, more debtors, illegible uploads: find the next failure before a user does.



Deeper evidence

OCR for scanned statements; invoice-level creditor reconciliation; cross-document conflict detection.



Beyond working capital

Delayed receivables, compliance notices, supplier disputes: the same investigate-then-prove pattern, problem-first.

DELIBERATELY OUT OF SCOPE FOR V0.7.0

TReDS and banking API integrations, autonomous actions on the owner's behalf, and multi-agent orchestration. KARBHARI investigates and recommends; the owner decides and acts.

READ EVERY DOCUMENT • CROSS-CHECK EVERY NUMBER • PROVE EVERY RUPEE

KARBHARI

WORKING CAPITAL GUARDIAN

Find what your numbers are hiding.